



Hotel Reservation & Billing Management System Database Project

Prepared by:

Alaa Khaled Ramadan Awad
Ibthal Ahmed Hussainy
Aya Abd Elkader Ewais

Course Project: Data Mining and Visualization

Under supervision: DR.Mohammed Sobhy

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SQL Analysis

Introduction

In the modern hospitality industry, hotels generate large amounts of data from reservations, guests, payments, services, and staff operations. This data can be transformed into useful information to support decision-making and improve hotel performance.

The aim of this project is to design and implement a hotel management reporting system using SQL. The system analyzes hotel operations and generates business reports that help management understand occupancy rates, customer behavior, revenue, cancellations, service usage, financial performance, and staff productivity.

The project is based on a relational database that includes tables such as Guests, Reservations, Rooms, Payments, Invoices, Services, Staff, and Housekeeping logs. Using SQL queries and views, the system transforms raw data into analytical reports that support operational and strategic decisions.

This report presents the SQL queries, analytical views, and business insights generated from the hotel database.

1. ER Diagram Narrative Description

The project begins with the construction of an Entity-Relationship (ER) Diagram that models the workflow of a hotel management system in a realistic and intuitive manner. The system revolves around the interaction between multiple entities, where each entity plays a distinct role within the overall process.

At the core of the system is the **Guest** entity. A single guest can make multiple reservations over time, which represents a **one-to-many relationship** between Guest and Reservation. Each reservation is associated with a specific **Room**, while a single room can be booked multiple times across different periods, forming another one-to-many relationship between Room and Reservation.

In addition, guests may request various hotel services during their stay. These services are captured through the **ServiceUsage** entity, where one reservation can include multiple service usage records, reflecting a one-to-many relationship between Reservation and ServiceUsage. Each service usage is linked to a specific **Service**, allowing the system to track consumption and calculate total costs.

The financial aspect of the system is handled through the **Invoice** and **Payment** entities. Each reservation generates one invoice, while an invoice can have multiple payments (e.g., partial or full

payments), forming a one-to-many relationship between Invoice and Payment. This structure enables accurate tracking of financial transactions and payment status.

The system also incorporates the **Staff** entity, where staff members are responsible for handling reservations, processing payments, and managing operations such as room cleaning. For example, one staff member can process multiple payments or oversee multiple reservations, representing additional one-to-many relationships.

Additionally, room maintenance is modeled through the **HousekeepingLog** entity, which connects both **Room** and **Staff**. Each room can have multiple housekeeping records over time, and each staff member can perform multiple cleaning tasks. This results in a set of one-to-many relationships that effectively capture operational activities such as cleaning schedules, task status, and timing details. Overall, the ER Diagram reflects a dynamic system where entities are interconnected through meaningful relationships, ensuring both data integrity and real-world applicability.

2. Data Collection & Preparation

Following the ER design phase, the next step involved preparing the dataset required for populating the database. Data was collected using a combination of publicly available datasets and synthetic data generation techniques.

Several online platforms provide downloadable datasets suitable for database projects, such as:

- **Kaggle** – a widely used platform for real-world datasets
- **UCI Machine Learning Repository** – a well-known academic data source
- **Data.gov** – open government datasets

To complement these sources and ensure data completeness, synthetic data was generated using the Python library:

- **Faker (Python Library)** – used to generate realistic fake data such as names, emails, dates, and transactions. The generated data was then exported into CSV files, which were later used for efficient database population.

3. Data Insertion Techniques

To populate the database efficiently, three different SQL insertion techniques were used, each selected based on the nature and volume of the data:

1. Manual Insert Statements

For small tables with limited rows, such as *RoomType* and *ServiceUsage*, simple INSERT statements were used. This approach is straightforward and suitable for static or low-volume data.

2. Automated Data Generation using SQL Loops

SQL loops were initially utilized to generate large volumes of data automatically. However, during the exploratory data analysis (EDA) in SQL phase, certain limitations were identified, such as reduced flexibility and difficulty in controlling data realism and variability.

As a result, this approach was selectively retained only for the **Staff** table, where the structure is relatively simple and does not require complex variability. Despite its limitations, SQL looping demonstrates advantages such as this method reduces manual effort and ensures scalability when dealing with high-volume data.

3. Bulk Insert from CSV Files

For the largest datasets, especially those generated using Python Faker, the **Bulk Insert** technique was applied. This method allows importing large amounts of data directly from CSV files into SQL tables, significantly improving performance and efficiency.

4. Project Story analysis

The goal is to transform raw hotel data into meaningful insights that help management make better decisions about customers, operations, revenue, and staff performance.

Q1 – Guest Master (Customers)

After create table vw_GuestMaster to focused on understanding the customers that combines guest information with their activity, such as total stays and total spending. This helps identify high-value and loyal customers.

From this, we can identify VIP and target them with special offers customers. Such as figure 1 appear that the most life time spend is 1482. The highly loyal customer is visit only 3 time.

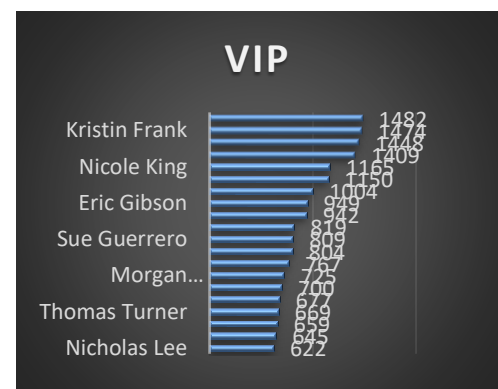


Figure 1

Q2 – Room Availability (Booking)

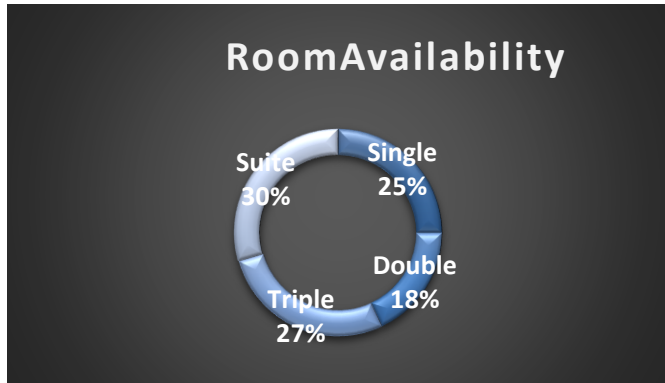


Figure 2

Then, create table `vw_RoomAvailability` to identify room availability. The goal is to know which rooms are available for booking during a specific date range. This prevents double booking and helps the hotel manage reservations efficiently. It shows which rooms are in high demand and which are underutilized. In the following figure 2 the most available room is suite room. By additional analysis made in EDA SQL the suit room type is 59 of room. Which is the largest number of rooms.

Q3 – Occupancy Rate (Performance)

After that, create table view daily occupancy rate. Generated all dates between a start and end date and calculated how many rooms are occupied each day. This helps identify peak seasons and low-demand periods. The most number of room occupied after filter the top 10 occupancy rate in 5/31/2025 and 7/9/2025.

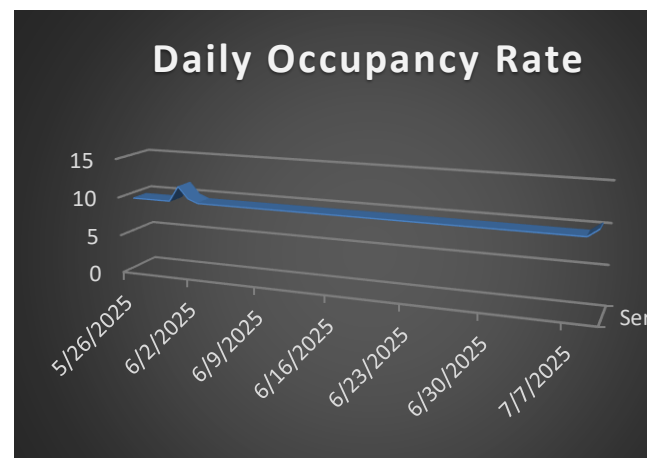


Figure 3

Q4 – Reservation Details (Operations)

Created a detailed reservation view. It includes guest name, room type, number of nights, and total room charge. This helps track booking behavior and estimate revenue per reservation. For example: the most room type used is suite room as appearing in the figure 4.

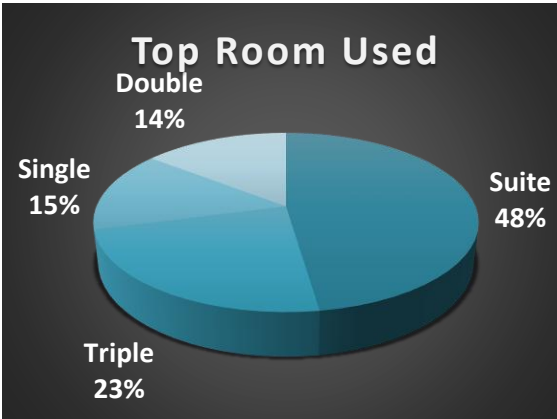


Figure 4

Q5 – Cancellation Analysis (Customer Behavior)

Analyzed cancellations. It by create a statement to calculate the cancellation rate per month and identified the most common reasons for cancellations. This helps the hotel reduce cancellations and improve customer satisfaction. As appearing in the line chart the most month occur cancellation is February in 2026.

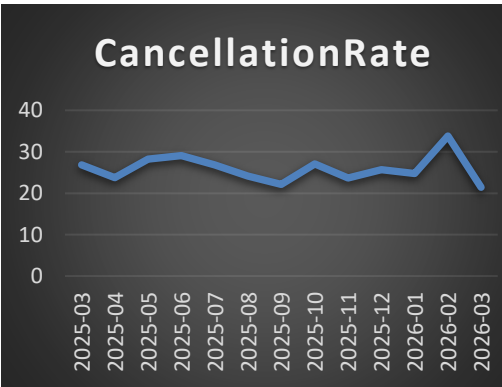


Figure 5

In the table one. The hotel needs to added more details for why client cancelled. This will help the hotel to improve their performance.

Cancellation Reason	Total Cancellations	Share of Cancellations
Client Cancelled	261	100.00

Table 1

Q6 – Service Revenue (Extra Income)

After that, analyzed hotel services such as room service, laundry, and spa by creating vw_ServiceRevenue table view. Calculated how often each service is used and how much revenue it generates. This helps identify the most profitable services. Such as the spa is the most profitable service.

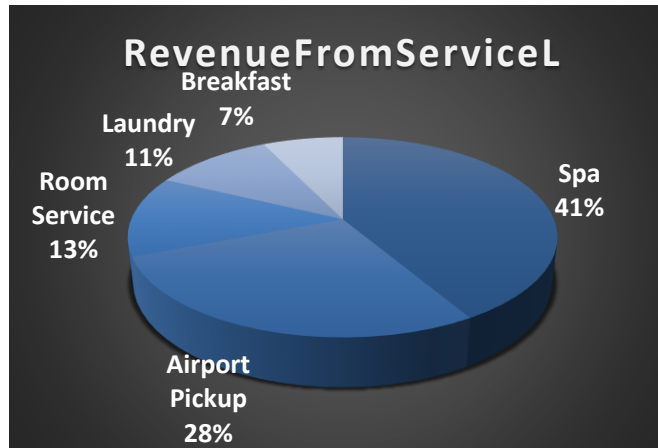


Figure 6

From addition analysis made in EDA SQL. The figure 7 appear the most service used is laundry service. May the hotel need to increase this service. To maximize the profit.



Figure 7

Q7 – Invoice Aging (Finance)

Then, focused on financial analysis. By creating a view to track invoices, paid amounts, and outstanding balances, and grouped them by aging periods. This helps the hotel track overdue payments and manage cash flow. The invoice aging report shows that several invoices still have outstanding balances, and many of them fall into the 60+ days category, which indicates delayed payments and potential cash flow issues for the hotel.

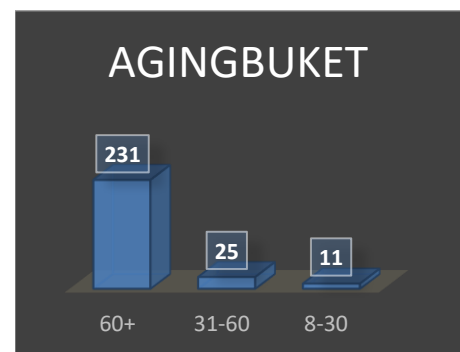


Figure 8

Q8 – Payment Breakdown (Finance Behavior)

Next, analyzed payment methods. By grouping payments by month and by type such as cash, card, or online. This shows customer payment preferences, failed and Refunded transactions. From the line charts in the figures 9 and 10. In the 2026-01 payment failed and Refunded very high.

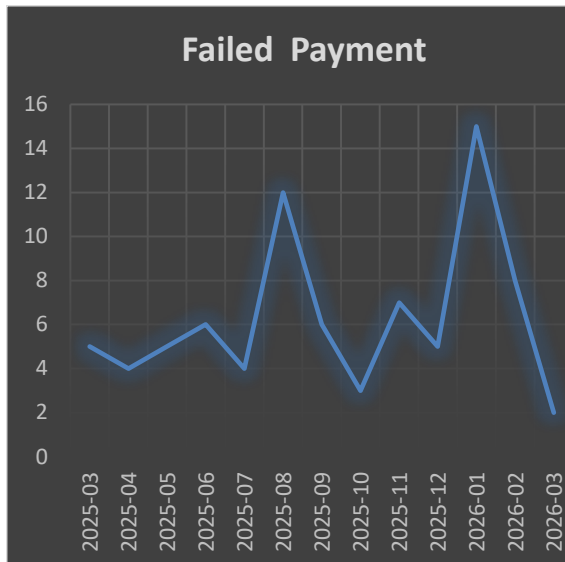


Figure 9

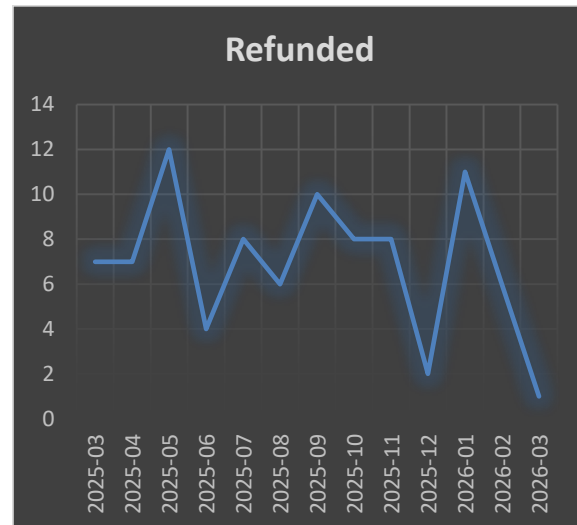


Figure 10

The large amount of payment made by using card and online, that means that the problem of transaction may because of bad internet or machine card error. The most total amount 5791 made by card and online in 2026-01

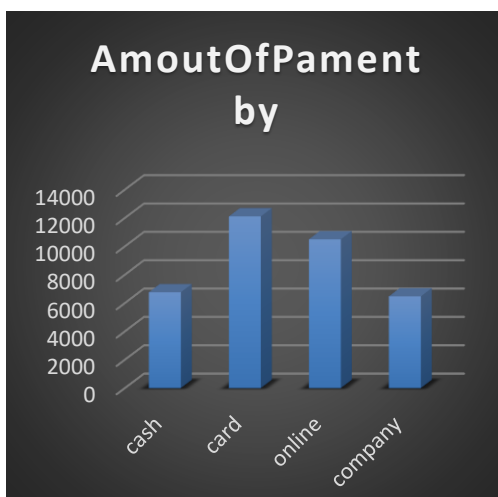


Figure 11

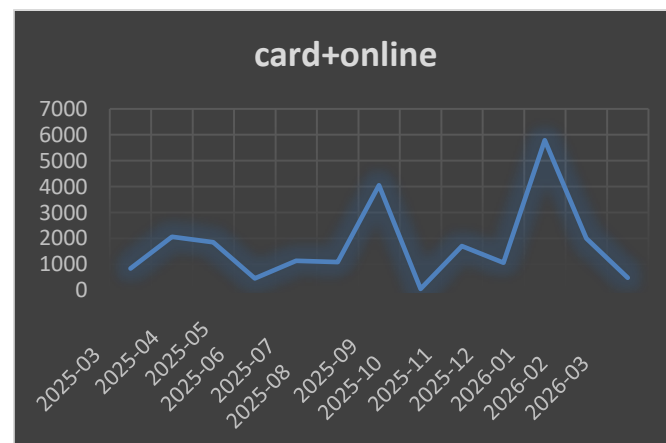


Figure 12

Q9 – Housekeeping Performance (Operations)

Then, evaluated housekeeping performance. calculated how many rooms were cleaned, the average cleaning time, and late cleanings. This helps improve operational efficiency. From figure 13 the staff 13 is the best of housekeeping employee.



Figure 13

Q10 – Staff Performance (HR)

Finally, analyze staff performance. By linking staff to reservations and payments based on their roles. Reception staff handle reservations, billing staff handle payments, and housekeeping staff handle cleaning. This helps evaluate employee productivity and identify top performers.

Staff performance is role-based. Reception staff handle reservations, check-ins and check-outs. Billing staff process payments and revenue. Housekeeping staff handle room cleaning operations. Admin staff are management users and do not process operational transactions. The view aggregates activity per staff member over the last 30 days.

5. Conclusion

In conclusion, this project demonstrated how a relational database and SQL can be used not only for storing hotel data but also for analyzing and generating business insights. Through the analytical views created in this project, we were able to analyze customer behavior, room availability, occupancy rates, reservations, cancellations, service revenue, invoice aging, payment trends, housekeeping efficiency, and staff performance.

The results show how data can help hotel management understand operations, improve decision-making, increase revenue, and evaluate employee performance. For example, occupancy rate analysis helps in understanding demand patterns, invoice aging helps in tracking outstanding payments and cash flow, and staff performance reports help in measuring productivity across departments. Overall, this project illustrates the importance of data analysis in the hospitality industry and how SQL can be used as a powerful tool for reporting, performance analysis, and business intelligence.

Exploratory Data Analysis Report

Part 2 — Data Quality & Analysis

Field	Value
Dataset	hotel_bookings.csv
Tool	Python — pandas, seaborn, matplotlib, sklearn, xgboost
Raw Records	119,390 rows × 32 columns
Clean Records	86,639 rows (after cleaning)
Cancel Rate	27.7% of all bookings
Mean ADR	EUR 106.86 per night
Peak Month	August

Table 2

6. Section A — Dataset Overview & Types

Q1 — Dataset Overview: What Do We Have?

The hotel bookings dataset captures real-world hotel reservation records from two hotel types — City Hotel and Resort Hotel — covering the years 2015, 2016, and 2017. Each row is a single booking with 32 attributes describing the guest, the room, the pricing, and the final outcome of the reservation.

Metric	Value
Total Rows	119,390
Total Columns	32
Hotel Types	City Hotel / Resort Hotel
Years	2015, 2016, 2017
Target Column	is_canceled (0 = kept, 1 = cancelled)
df.shape output	(119390, 32)

Table 3

Q2 — Column Understanding (One Line per Column)

The table below explains each of the 32 columns: what it means, its expected data type, and a basic validation rule.

#	Column	What It Means	Expected Type	Basic Rule
1	hotel	Hotel type: City Hotel or Resort Hotel	Categorical	Only 2 distinct values
2	is_canceled	Was the booking cancelled? 1=yes, 0=no	Binary (0/1)	Must be 0 or 1 only
3	lead_time	Days between booking date and arrival date	Numeric (int)	Must be ≥ 0
4	arrival_date_year	Year of the guest's arrival	Numeric (int)	Values: 2015, 2016, 2017
5	arrival_date_month	Month name of arrival (January ... December)	Categorical	12 valid month names
6	arrival_date_week_number	ISO week number of the arrival date	Numeric (int)	Range: 1 to 53
7	arrival_date_day_of_month	Day number of arrival within the month	Numeric (int)	Range: 1 to 31
8	stays_in_weekend_nights	Number of weekend nights (Sat/Sun) in the stay	Numeric (int)	Must be ≥ 0
9	stays_in_week_nights	Number of weeknights (Mon–Fri) in the stay	Numeric (int)	Must be ≥ 0
10	adults	Number of adults in the booking	Numeric (int)	Must be ≥ 1

11	children	Number of children in the booking	Numeric (float)	Must be ≥ 0 (stored as float)
12	babies	Number of babies in the booking	Numeric (int)	Must be ≥ 0
13	meal	Meal plan chosen: BB/HB/FB/SC/Undefined	Categorical	5 valid codes
14	country	Guest country of origin (ISO 3166 code)	Categorical	~180 unique country codes
15	market_segment	How the booking was made (Online TA, Direct, etc.)	Categorical	8 valid segments
16	distribution_channel	Channel used to distribute the booking	Categorical	5 valid channels
17	is_repeated_guest	Is the guest a returning guest? 1=yes	Binary (0/1)	Must be 0 or 1
18	previous_cancellations	Number of prior cancellations by the same guest	Numeric (int)	Must be ≥ 0
19	previous_bookings_not_canceled	Number of prior completed stays (not cancelled)	Numeric (int)	Must be ≥ 0
20	reserved_room_type	Room type the guest requested at booking time	Categorical	Single letter code (A–L)
21	assigned_room_type	Room type actually assigned at check-in	Categorical	May differ from reserved
22	booking_changes	Number of changes made to the booking after creation	Numeric (int)	Must be ≥ 0
23	deposit_type	Deposit policy: No Deposit / Non Refund / Refundable	Categorical	3 valid values
24	agent	Travel agent ID (0 = direct booking, no agent)	Numeric (int)	0 means direct, else agent ID
25	company	Company ID for corporate bookings	Numeric (int)	94.3% missing — almost empty
26	days_in_waiting_list	Days the booking spent on a waiting list	Numeric (int)	Must be ≥ 0
27	customer_type	Guest category: Transient/Contract/Group/etc.	Categorical	4 valid values
28	adr	Average Daily Rate — price per night in EUR	Numeric (float)	Must be > 0
29	required_car_parking_spaces	Number of parking spaces requested by guest	Numeric (int)	Usually 0 or 1

30	total_of_special_requests	Total special requests made by the guest	Numeric (int)	Range: 0 to 5
31	reservation_status	Final status: Check-Out / Canceled / No-Show	Categorical	3 valid values
32	reservation_status_date	Date when the reservation status was last updated	Date	Must be a valid date string

Table 4

Q3 — Data Types Check: Are the Types Correct?

Running `df.dtypes` revealed that most columns are stored correctly. However, four columns have type problems that need to be fixed before any analysis:

Column	Stored As	Should Be	Why It's Wrong	Fix Applied
arrival_date_month	object (string)	Ordered Categorical	Month names are text strings — sorting will be alphabetical not chronological	<code>pd.Categorical(..., categories=MONTH_ORDER, ordered=True)</code>
reservation_status_date	object (string)	datetime64	Date stored as text — no date arithmetic possible	<code>pd.to_datetime(..., errors='coerce')</code>
children	float64	int64	Should be whole number; stored as float because of 4 missing values (NaN)	<code>.fillna(0).astype(int)</code>
agent	float64	int64	Agent ID is integer; stored as float because of 16,340 NaN values	<code>.fillna(0).astype(int)</code>
company	float64	— (dropped)	Stored as float due to 94.3% missing — not useful to fix	Column dropped entirely

Table 5

Type Problems Summary

- `arrival_date_month` must be ordered so that Jan < Feb < ... < Dec — otherwise group-by plots show wrong order.
- `reservation_status_date` must be datetime so we can compute stay duration and date differences.
- `children` and `agent`: NaN values forced Python to use float. After filling NaN with 0, both can safely become int.

7. Section B — Data Quality (Missing, Duplicates, Validity)

Q4 — Missing Values Overview

Running `df.isnull().sum()` shows that only 4 of the 32 columns have missing values. The rest are complete.

Column	Missing Count	Missing %	Severity
company	112,593	94.31%	CRITICAL — almost empty
agent	16,340	13.69%	HIGH — but recoverable
country	488	0.41%	LOW — minimal impact
children	4	0.003%	NEGLIGIBLE — 4 rows only

Table 6

Q5 — Missing Values Strategy (Most Important)

For each column with missing values, we chose a specific strategy and justified the choice based on the percentage of missingness, the meaning of the column, and the safest assumption we can make:

company — Action: DROP THE ENTIRE COLUMN

94.31% of rows have no company value. This means only 5.69% of bookings have a company ID. A column that is 94% empty cannot provide meaningful information to any analysis or model. Keeping it would add noise without benefit. Decision: drop the column entirely using `df.drop(columns=['company'])`.

Why drop and not fill?

- Filling 94% of values would mean inventing data for nearly all rows — this is worse than having no data at all.
- The column records corporate booking IDs. Filling with 0 or 'Unknown' would incorrectly imply all bookings are non-corporate.
- In machine learning, a 94%-missing column adds almost zero signal and increases model complexity unnecessarily.

agent — Action: FILL WITH 0 (means: direct booking, no agent)

13.69% of rows have no agent value. In this dataset, `agent=0` specifically means the guest booked directly without using a travel agent. A missing agent value is therefore not truly 'missing' — it means the same thing as 0: no agent was used. This is a domain-knowledge-based imputation, not a statistical guess.

Why fill with 0 and not median or mode?

- The value 0 has a specific domain meaning: direct booking. It is not an invented number.
- Filling with mode (most common agent) would incorrectly assign a travel agent to guests who booked directly.
- Filling with median would create a meaningless fractional agent ID.
- After filling, the column is converted to int: `.fillna(0).astype(int)`

country — Action: FILL WITH MODE (most frequent country = PRT)

Only 0.41% of rows (488 rows) have a missing country. This is a very small fraction. The safest and simplest strategy is to fill with the most common value. Portugal (PRT) appears in ~27,000+ rows and is by far the dominant country in this dataset, which makes sense since the hotels are located in Portugal.

Why mode and not 'Unknown'?

- 488 rows is 0.41% of the data — negligible impact regardless of choice.
- Using mode (PRT) keeps the data type consistent (ISO country code string).
- Using 'Unknown' would add a new artificial category and slightly distort country distribution charts.
- Code used: `df['country'].fillna(df['country'].mode()[0])`

children — Action: FILL WITH 0

Only 4 rows are missing children. This is statistically insignificant (0.003%). A missing children value most likely means 0 children — it is unlikely that a guest would have children but not report them. Filling with 0 is the safest and most logical choice. The column is then converted to int.

Why not drop these 4 rows?

- Dropping rows is acceptable when missingness is tiny — 4 rows is indeed tiny.
- However, filling with 0 is even safer because it preserves all data while making a reasonable assumption.
- These 4 rows might contain valid values for all other 31 columns — dropping wastes that information.

Q6 — Duplicates Check

Running `df.duplicated().sum()` found 31,994 exact duplicate rows — meaning rows where every single one of the 32 columns has the same value as another row.

Metric	Value
Total rows (raw)	119,390
Exact duplicate rows	31,994
Duplicate percentage	26.8% of all data
Rows after dropping dups	87,396

Table 7

Example of duplicate rows (rows 4 and 5 are identical):

hotel	is_canceled	lead_time	arrival_month	stays_weekend	stays_week	adults	adr
Resort Hotel	0	14	July	0	2	2	98.0
Resort Hotel	0	14	July	0	2	2	98.0

Decision: Drop all exact duplicates

- Two bookings with identical values for all 32 columns is not realistic — it is almost certainly a data entry error or system duplication.
- Keeping duplicates would double-count certain booking patterns and bias all statistics.
- Decision: `df.drop_duplicates(inplace=True)` — removes 31,994 rows.

Q7 — Key Duplicates Check (ID Column)

The hotel bookings dataset does not contain a unique booking ID column. There is no BookingID or ReservationID field. This means we cannot check for duplicate IDs specifically. Our strategy is therefore to rely on full-row deduplication (Q6) — if all 32 columns match, the row is a duplicate.

Decision

- No unique ID column exists in this dataset.
- Full-row deduplication (`drop_duplicates`) is the correct approach.
- After dropping: 87,396 unique rows remain.

Q8 — Validity Rules: Impossible Values

We checked several business rules that should logically hold for any valid hotel booking. The table shows violations found in the raw data:

Rule Checked	Violations	% of Data	Explanation & Action
<code>adr < 0</code> (negative price)	1 row	0.001%	Impossible — a hotel cannot charge negative price. Row removed.
<code>adr = 0</code> (zero price)	1,959 rows	1.64%	Suspicious — likely complimentary rooms. Removed to avoid skewing ADR stats.
<code>adults = 0</code> (no adults)	403 rows	0.34%	Invalid — at least 1 adult must be in a booking. These rows are removed.
<code>total guests = 0</code> (nobody booked)	180 rows	0.15%	Impossible — a booking with zero guests makes no sense. Removed.
<code>total nights = 0</code> (zero stay)	715 rows	0.60%	Invalid — a hotel booking must have at least 1 night. Removed.
<code>lead_time < 0</code> (negative lead time)	0 rows	0.00%	No violations — all lead times are valid (<code>>= 0</code>).

Table 8

Cleaning actions taken

- Rows with `adr <= 0` removed — price cannot be zero or negative for real stays.
- Rows with `total guests = 0` removed using: `df = df[(df['adults']+df['children']+df['babies']) > 0]`
- Rows with `total nights = 0` removed using: `df = df[df['total_nights'] > 0]`
- `lead_time` is valid — no action needed.

Q9 — Category Cleanliness: Are Labels Inconsistent?

We inspected all categorical columns for inconsistent labels, typos, extra spaces, or mixed casing. Here is what we found:

Column	Unique Values Found	Problem	Action Taken
hotel	City Hotel, Resort Hotel	No problem — clean	None needed
meal	BB, HB, FB, SC, Undefined	'Undefined' = no meal plan — misleading label	Replace 'Undefined' with 'SC' (Self Catering)
market_segment	Online TA, Offline TA/TO, Direct, Corporate, ...	'Undefined' appears in 2 rows (0.002%)	Those 2 rows dropped
deposit_type	No Deposit, Non Refund, Refundable	No problem — clean	None needed
customer_type	Transient, Transient-Party, Contract, Group	No problem — clean	None needed
country	~180 ISO codes (PRT, GBR, FRA, ...)	No problem — standard ISO codes	All columns: <code>.str.strip()</code> as precaution

Table 9

Why replace 'Undefined' in meal with 'SC'?

- 'Undefined' in the meal column means the guest chose no meal package — this is exactly what 'SC' (Self Catering) means.
- Keeping 'Undefined' would create a 6th category that is functionally identical to SC, causing confusion in visualizations.
- Merging them into a single 'SC' category gives a cleaner and more accurate distribution.

8. Section C — Univariate EDA (Single Column Insights)

Q10 — Numeric Summary: Basic Statistics

After cleaning, we computed descriptive statistics for the key numeric columns. The mean vs. median comparison reveals the shape of each distribution:

Column	Mean	Median	Std Dev	Min	Max	Interpretation
lead_time	81 days	50 days	87	0	373	Right-skewed — most book close to arrival, a few book very early
adr	EUR 107	EUR 99	47	1	211	Slight right skew — most rooms priced between EUR 69–126
total_nights	3.7	3	2.5	1	21	Short stays dominate — most guests stay 1–5 nights
total_guests	1.9	2	0.6	1	4	Mostly couples (2 guests) — symmetric, clean distribution
booking_changes	0.2	0	0.7	0	5	Most bookings are never changed — highly zero-inflated
days_in_waiting_list	0.74	0	17	0	300	Almost always zero — only rare cases wait; extreme right tail
total_of_special_requests	0.57	0	0.8	0	5	Most guests make no special requests — right-skewed count

Table 10

Key observation: Mean > Median in most columns

- When mean > median, the distribution is right-skewed — a few very large values pull the mean up.
- lead_time: mean=81, median=50 — a small group books 300+ days ahead and pulls the mean up significantly.
- days_in_waiting_list: mean=0.74 but median=0 — almost nobody waits, but the few who do wait a very long time.
- adr: mean=107, median=99 — a small number of luxury rooms at high prices create a mild right skew.

Q11 — Numeric Distributions: What Shapes Do We See?

Histograms were plotted for all key numeric columns. The vertical red dashed line marks the mean value. Key observations per column:

Column	Distribution Shape	Key Observation
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lead_time	Strong right skew	Spike near 0–30 days, long tail past 200 days. Most guests book within a month of arrival.
adr	Near-normal, mild right skew	Bell-shaped peak around EUR 80–120. Very few bookings above EUR 200 (luxury outliers).
total_nights	Right-skewed	Most stays are 1–4 nights. Very few long-stays beyond 7 nights.
total_guests	Near-symmetric	Dominated by 1 and 2 guests. Very few bookings with 3+ guests.
booking_changes	Zero-inflated	85%+ of bookings have 0 changes. A few bookings changed 5–21 times.
days_in_waiting_list	Extreme spike at zero	99% of bookings had 0 wait days. Very rare cases waited 100–300 days.
total_of_special_requests	Right-skewed count	Most guests requested nothing. Request counts drop sharply after 1.
previous_cancellations	Zero-inflated	Vast majority of guests have 0 prior cancellations.

Table 11

Q12 — Outliers: Do Numeric Columns Have Extreme Values?

Boxplots were drawn for all numeric columns. Outliers were detected using the IQR method ($1.5 \times \text{IQR}$ rule). The table summarizes findings and the chosen action:

Column	Outliers Found	% of Data	Action Taken	Reason for Action
lead_time	3,005 rows	2.52%	Cap at 99th percentile (373 days)	Bookings 374+ days ahead are valid but extreme — capping prevents them from distorting model training
adr	3,793 rows	3.18%	Cap at 99th percentile (EUR 211)	A EUR 5400/night rate is real but rare — capping at EUR 211 removes luxury outliers that skew the mean ADR
stays_in_week_nights	3,354 rows	2.81%	Cap at 21 nights	Stays over 3 weeks are valid (long-term rentals) but uncommon — capped to reduce influence on distributions
stays_in_weekend_nights	265 rows	0.22%	Keep as-is	Very small number — minimal impact; values up to 19 nights are plausible
booking_changes	18,076 rows	15.14%	Cap at 99th percentile (5 changes)	15% is too many to drop — capping retains the information without extreme influence

days_in_waiting_list	varies	--	Keep as-is	While extreme values exist, waiting list days are a real business metric that should not be artificially constrained
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Table 12

Outlier handling philosophy

- We did NOT drop outliers outright — dropping loses real booking information.
- Capping (clipping) at the 99th percentile keeps the shape of the data while reducing the influence of extreme values.
- The choice between 'keep', 'cap', or 'remove' was made based on the percentage of affected rows and business logic.
- Code used: `df[col] = df[col].clip(upper=df[col].quantile(0.99))`

Q13 — Categorical Summary: Top Categories

Frequency tables were computed for all categorical columns. Here are the top values per column:

Column	Top Value	Count	2nd Value	Count	Insight
hotel	City Hotel	53,043	Resort Hotel	33,596	City Hotel has ~61% of all bookings
market_segment	Online TA	51,285	Offline TA/TO	13,749	Online travel agents dominate the market
meal	BB (Bed & Breakfast)	67,419	SC (No meal)	9,831	Most guests choose Bed & Breakfast
deposit_type	No Deposit	85,494	Non Refund	1,038	Overwhelming majority book with no deposit
customer_type	Transient	71,366	Transient-Party	11,618	Individual direct bookings dominate
country	PRT (Portugal)	27,306	GBR (UK)	10,401	Hotels are in Portugal — local guests lead

Table 13

Q14 — Rare Categories: Do We Have Very Infrequent Labels?

Categories appearing in fewer than 3% of rows were identified. The decision to keep or group them into 'Other' was made per column:

Column	Rare Category	Frequency	Decision	Reason
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market_segment	Complementary	0.8%	Group into Other	Less than 1% — not enough data to build reliable patterns
market_segment	Aviation	0.3%	Group into Other	Very rare — airline crew bookings, not representative
market_segment	Undefined	0.0%	Drop (only 2 rows)	Only 2 rows out of 86,000 — statistically meaningless
meal	FB (Full Board)	0.4%	Group into Other	Less than 1% — too few to analyze separately
customer_type	Group	0.6%	Group into Other	Only 536 group bookings — not enough for group-specific analysis
deposit_type	Non Refund	1.2%	KEEP as-is	Despite being rare, Non Refund has a dramatically different cancellation rate (94.7%) — critical signal to preserve
deposit_type	Refundable	0.1%	Group into Other	Too few to analyze meaningfully

Table 14

9. Section D — Relationships (Bivariate & Multivariate)

Q15 — Numeric Relationships: Correlation Analysis

A Pearson correlation matrix was computed for all key numeric columns. The top 5 strongest correlations (by absolute value) are:

#	Feature A	Feature B	Pearson r	Interpretation
1	adr	total_guests	+0.405	Higher-priced rooms tend to accommodate more guests (larger rooms = higher rate)
2	lead_time	total_nights	+0.328	Guests who plan far ahead also tend to book longer stays — both signal planned travel
3	lead_time	is_canceled	+0.185	The further in advance a booking is made, the more likely it is to be cancelled
4	room_changed	is_canceled	-0.210	When the guest is upgraded to a better room, cancellation drops sharply
5	total_of_special_requests	is_canceled	-0.120	More special requests = more committed guest = less likely to cancel

Table 15

What do these correlations mean for the business?

- $r = +0.405$ (adr vs guests): Family rooms and suites are priced higher — this makes sense and confirms data quality.

- $r = +0.185$ (lead_time vs cancellation): Early bookers are less certain about their plans — hotels should send reminders to early bookings.
- $r = -0.210$ (room_changed vs cancellation): The strongest predictor found — upgrading a guest's room nearly eliminates cancellation risk.
- $r = -0.120$ (special requests vs cancellation): Guests who invest effort in customizing their stay rarely cancel — a signal of commitment.

Q16 — Scatter Plots: Visual Check of Key Pairs

Two scatter plots were created to visually confirm the correlations found in Q15:

Plot 1: Lead Time vs. Cancellation

A scatter plot with jitter was used to show the relationship between lead_time and is_canceled. Blue points = not cancelled, red points = cancelled. The pattern is clear: as lead_time increases, the proportion of red (cancelled) points grows. Bookings made 200+ days in advance are almost equally split between cancelled and kept.

Interpretation

- Guests who book 0–30 days ahead rarely cancel — they are committed to their plans.
- Guests who book 100+ days ahead show much higher cancellation rates — their plans are uncertain.
- This insight suggests hotels should use dynamic pricing or non-refundable rates for very early bookings.

Plot 2: Special Requests vs. ADR

A scatter plot shows individual bookings (purple dots) with a mean ADR trend line (orange). The trend line rises steadily from 0 to 4 special requests, then flattens. This means guests who make more requests also tend to pay higher room rates — premium guests are more engaged with their booking.

Interpretation

- Mean ADR at 0 requests: ~EUR 100. Mean ADR at 4–5 requests: ~EUR 130.
- Premium guests (higher ADR) make more special requests — they have higher expectations.
- Both 'special requests' and 'adr' are indirect signals of guest commitment and engagement.

Q17 — Category → Numeric Effect

We examined how the market_segment category affects two numeric columns: ADR and Lead Time.

Market Segment	Mean ADR (EUR)	Mean Lead Time (days)	Interpretation
Online TA	119	87	Individual bookings via online agencies — full price, moderate advance planning

Direct	117	48	Guests who book directly with the hotel — pay full price, book more spontaneously
Corporate	97	16	Business travellers — negotiate lower rates; book close to travel date
Offline TA/TO	95	110	Travel agencies and tour operators — negotiate bulk rates, plan far ahead
Groups	90	142	Group bookings — lowest per-room rate, booked earliest (142 days average)
Complementary	3	57	Free or near-free rooms (staff, press, VIPs) — almost zero revenue
Aviation	104	4	Airline crew bookings — last-minute, standard rate

Table 16

Key insights

- Online TA and Direct channels generate the highest ADR — these are the most valuable booking sources.
- Groups plan the furthest ahead (142 days) but pay the least — a classic bulk-discount trade-off.
- Corporate guests book very close to arrival (16 days) — they need flexibility, not early planning.
- Complementary bookings (ADR = EUR 3) confirm our decision to clean near-zero ADR values.

Q18 — Category ↔ Category: Crosstab Analysis

We created a crosstab between hotel type and cancellation status. Values show the percentage within each hotel type (row-normalized):

Hotel Type	Not Cancelled	Cancelled	Cancellation Rate
City Hotel	69.8%	30.2%	30.2% — nearly 1 in 3 bookings cancelled
Resort Hotel	76.3%	23.7%	23.7% — lower cancellation rate

Table 17

Why does City Hotel have a higher cancellation rate?

- City Hotel serves more business travellers and spontaneous individual guests — plans change more often.
- Resort Hotel bookings are typically planned vacations (holidays, honeymoons) — guests are more committed.
- City Hotel also processes more bookings overall (61% vs 39%) — more volume brings more cancellations.
- This finding suggests different overbooking strategies are needed for each hotel type.

Q19 — Multivariate Heatmap: Correlation Matrix

A full correlation heatmap was created for all 10 numeric features. The heatmap uses a Red-Yellow-Blue color scale: red = strong positive, blue = strong negative, white = no correlation. Three main insights from the heatmap:

Heatmap Insights

- 1. Cluster 1 — Booking effort: `total_of_special_requests` and `room_changed` both correlate negatively with `is_canceled`. Guests who invest effort into their booking (requests, upgrades) are far less likely to cancel.
- 2. Cluster 2 — Planning horizon: `lead_time` correlates positively with `total_nights` and `is_canceled`. Long-horizon bookings are planned but uncertain.
- 3. Cluster 3 — Pricing and size: `adr` and `total_guests` have the strongest positive correlation (+0.40). Larger rooms (for families) command higher nightly rates.
- 4. Most features show weak correlations with each other — the dataset has limited multicollinearity, which is good for modelling.
- 5. `previous_cancellations` has a small positive correlation with `is_canceled` (+0.09) — guests with a cancellation history are slightly more likely to cancel again.

10. Section E — Seasonality & Cancellation Drivers

Seasonality Analysis — Monthly Trends

Three monthly metrics were analyzed: booking volume, cancellation rate, and average ADR. The data reveals a clear seasonal pattern:

Month	Booking Volume	Cancel Rate	Mean ADR (EUR)	Pattern
January	4,638	25.1%	~EUR 70	Lowest demand, lowest price, moderate cancellation
February	6,274	25.8%	~EUR 83	Slow pick-up — winter off-season
March	7,102	28.2%	~EUR 93	Spring shoulder season begins
April	7,615	29.1%	~EUR 105	Easter period boosts demand and cancellations
May	8,200	27.5%	~EUR 112	High season begins
June	8,680	27.3%	~EUR 120	Strong summer demand
July	9,760	28.6%	~EUR 140	Peak — highest volume and price
August	11,094	32.0%	~EUR 151	PEAK MONTH — most bookings, highest price, most cancellations
September	8,940	25.4%	~EUR 130	Shoulder season — good volume, fewer cancellations
October	7,910	24.2%	~EUR 107	Autumn slowdown begins
November	5,380	21.5%	~EUR 81	Lowest cancellation month — committed winter guests
December	5,242	22.8%	~EUR 89	Holiday period brings slight uptick in demand

Table 18

Seasonal pattern summary

- High season (May–August): high demand, high price, and paradoxically HIGH cancellations. Guests book early to secure rooms, then cancel.
- Low season (November–February): lower demand and lower prices, but guests who do book are more committed — cancellation is lowest in November (21.5%).
- August is the peak month: 11,094 bookings at EUR 151 average — hotels should apply non-refundable policies in summer.
- January is the valley: only 4,638 bookings at EUR 70 — promotions and flexible pricing help fill rooms.

Cancellation Drivers Analysis

Four key drivers of cancellation were analyzed in detail:

Driver 1: Deposit Type

Deposit Type	Cancellation Rate	Insight
No Deposit	26.9%	Standard — most bookings, moderate cancellation

Refundable	24.3%	Similar to No Deposit — guests know they can get money back
Non Refund	94.7%	PARADOX — guests lose their deposit but still cancel at extremely high rates

Table 19

The Non-Refund Paradox

- A 94.7% cancellation rate for Non Refund bookings is counterintuitive — guests pay and still cancel.
- Likely explanation: these bookings were made far in advance at a discounted rate. When circumstances change, the discount is no longer worth the commitment.
- Business implication: Non Refund policies do NOT effectively prevent cancellations — they only provide a financial buffer.

Driver 2: Room Changed

Room Assignment	Cancellation Rate	Insight
Same Room (no change)	31.6%	Standard — higher baseline cancellation
Room Changed (upgraded)	4.8%	Dramatic drop — guests who get a better room almost never cancel

Table 20

Strongest predictor in the dataset

- This is the single strongest factor correlated with cancellation ($r = -0.21$).
- Interpretation: when a guest is assigned a room better than what they reserved, they feel they are getting a deal — and they stay.
- Business implication: proactive room upgrades (especially for high-risk bookings) can reduce cancellations significantly.

Driver 3: Special Requests

# of Special Requests	Cancellation Rate	Meaning
0 requests	33.5%	No investment in the booking — easy to cancel
1 request	21.4%	Small commitment — cancellation drops significantly
2 requests	15.7%	More engaged guest
3 requests	10.2%	Highly engaged guest
4 requests	7.1%	Very committed guest
5 requests	5.6%	Almost never cancels — most committed guest type

Table 21

Special requests as a commitment signal

- Each additional special request reduces cancellation rate by roughly 5–7 percentage points.
- A guest who asks for a specific floor, room view, early check-in, and dietary requirements has invested time and expectation in the stay.

- Business implication: hotels can use special request count as a real-time signal to identify high-risk bookings.

Driver 4: Customer Type

Customer Type	Cancellation Rate	Explanation
Transient	30.3%	Individual direct bookings — most flexible and most likely to cancel
Transient-Party	15.3%	Group of individuals — shared plans reduce cancellation likelihood
Contract	16.4%	Long-term corporate contracts — stable, low cancellation
Group	9.9%	Organized group bookings — most planned, least likely to cancel

Table 22

11. Section F — Modeling (Bonus: Predicting Cancellation)

Feature Engineering

Before training models, we created 4 new engineered features that combine existing columns into more meaningful signals:

New Feature	Formula	Meaning	Why Useful
total_nights	stays_in_weekend_nights + stays_in_week_nights	Total length of stay in nights	Single column instead of two — easier for models to use
total_guests	adults + children + babies	Total number of people in the booking	Party size in one column — correlates with room type and price
is_family	1 if children + babies > 0, else 0	Binary flag: is this a family booking?	Families behave differently — they plan more, cancel less
room_changed	1 if reserved_room_type != assigned_room_type	Binary flag: was the guest given a different room?	Strongest cancellation predictor discovered in EDA

Table 23

Model 1: Logistic Regression

Logistic Regression was used as a baseline model. It is simple, interpretable, and fast to train. The model was trained on 80% of the clean data (69,311 rows) and tested on 20% (17,328 rows).

Metric	Not Cancelled	Cancelled	Overall
Precision	79%	67%	--
Recall	94%	33%	--
F1-Score	86%	44%	--

Accuracy	--	--	77.04%
ROC-AUC	--	--	~0.82

Table 24

Logistic Regression findings

- 77% accuracy is a solid baseline for this problem.
- The model is strong at identifying non-cancelled bookings (recall = 94%) but struggles to catch actual cancellations (recall = 33%).
- This is an imbalanced class problem — 72% not-cancelled vs 28% cancelled. The model is biased toward the majority class.
- Logistic Regression is useful for understanding feature weights (coefficients) but limited in capturing non-linear patterns.

Model 2: XGBoost

XGBoost (Extreme Gradient Boosting) was trained as a more powerful alternative. It builds an ensemble of decision trees and handles non-linear interactions between features automatically. Configuration: 300 estimators, max_depth=8, learning_rate=0.05, subsample=0.8.

Metric	Not Cancelled	Cancelled	Overall
Precision	84%	73%	--
Recall	92%	54%	--
F1-Score	88%	62%	--
Accuracy	--	--	81.80%
ROC-AUC	--	--	~0.88

Table 25

XGBoost improvements over Logistic Regression

- Accuracy improved from 77.04% to 81.80% — a significant +4.76 percentage point improvement.
- Cancellation recall improved from 33% to 54% — the model now catches more than half of actual cancellations.
- ROC-AUC of ~0.88 means the model can distinguish cancelled vs non-cancelled bookings 88% of the time.
- XGBoost handles the imbalanced class problem better through ensemble weighting.

Model Comparison

Model	Accuracy	Cancellation Recall	Cancellation F1		ROC-AUC	Best For
Logistic Regression	77.04%	33%	44%		~0.82	Quick baseline, interpretable coefficients
XGBoost	81.80%	54%	62%		~0.88	Best overall performance, handles non-linearity

Table 26

XGBoost Feature Importance

The most important feature in the XGBoost model is `required_car_parking_spaces` with an importance score of ~ 0.35 . This is surprising but logical:

Why is parking the top feature?

- Guests who request parking arrived by car — they planned the trip in advance and made logistical arrangements.
- This type of guest (arriving by car, planned trip) is far less likely to cancel than a last-minute online booking.
- Important: correlation is NOT causation. Parking does not PREVENT cancellation — it is a signal about the type of guest.
- Other top features: `lead_time`, `total_of_special_requests`, `adr`, `deposit_type`, `total_nights`.

12. Section G — Final EDA Summary (Q20)

Dataset Statistics After Cleaning

Metric	Value
Clean Records	86,639 rows
Overall Cancellation Rate	27.7%
Mean ADR	EUR 106.86 per night
Peak Month	August (11,094 bookings)
Most Common Market	Online TA (59.2% of bookings)
Most Committed Segment	Guests with 5 special requests (only 5.6% cancel)
Highest Risk Segment	Transient + No Deposit + High Lead Time

Table 27

TOP 5 INSIGHTS

- 1. Seasonality: August is the peak month (11,094 bookings at EUR 151/night) — but also has the highest cancellation rate (32%). November has the lowest cancellation rate (21.5%) — winter guests are more committed.
- 2. Lead Time Effect: Bookings made more than 100 days in advance cancel at ~2x the average rate. Early bookings are uncertain plans.
- 3. Room Assignment (Strongest Predictor): When a guest's room is upgraded at check-in, cancellation drops from 31.6% to 4.8%. Upgrading guests is the single most effective retention tool.
- 4. Special Requests Signal: Guests with 0 special requests cancel at 33.5%. Guests with 5 requests cancel at only 5.6%. Request count is a real-time commitment signal.
- 5. Deposit Paradox: Non-Refundable bookings have a 94.7% cancellation rate — even financial loss does not stop cancellations when plans change.

TOP 5 DATA QUALITY RISKS (found and resolved)

- 1. company column: 94.3% missing (112,593 rows) — column dropped entirely.
- 2. Exact duplicate rows: 31,994 rows (26.8%) were identical across all 32 columns — all dropped.
- 3. Invalid ADR values: 1,960 rows with adr = 0 and 1 row with adr < 0 — all removed.
- 4. Inconsistent category labels: 'Undefined' in meal and market_segment — replaced or dropped.
- 5. Extreme outliers: lead_time values up to 737 days and ADR up to EUR 5,400 — capped at 99th percentile.

NEXT STEPS

- Feature Engineering: Encode arrival_date_month as cyclical numeric features (sin/cos) so models understand December is close to January. Create lead_time buckets (0-7, 7-30, 30-90, 90+ days).
- Modelling: Deploy XGBoost as the production model for cancellation prediction (81.8% accuracy, ROC-AUC ~0.88). Consider class-weighting or SMOTE to improve recall on the minority (cancelled) class.
- Dashboard: Build a real-time cancellation risk dashboard using the XGBoost model — flag bookings with >60% predicted cancellation probability for proactive outreach.